

Digital Signal Processing - Lecture 4

Negative frequencies, spectral leakage, and aliasing

Ewout Bergsma

Hanze University of Applied Sciences

Overview

This lecture

01 Negative frequencies

02 Spectral leakage

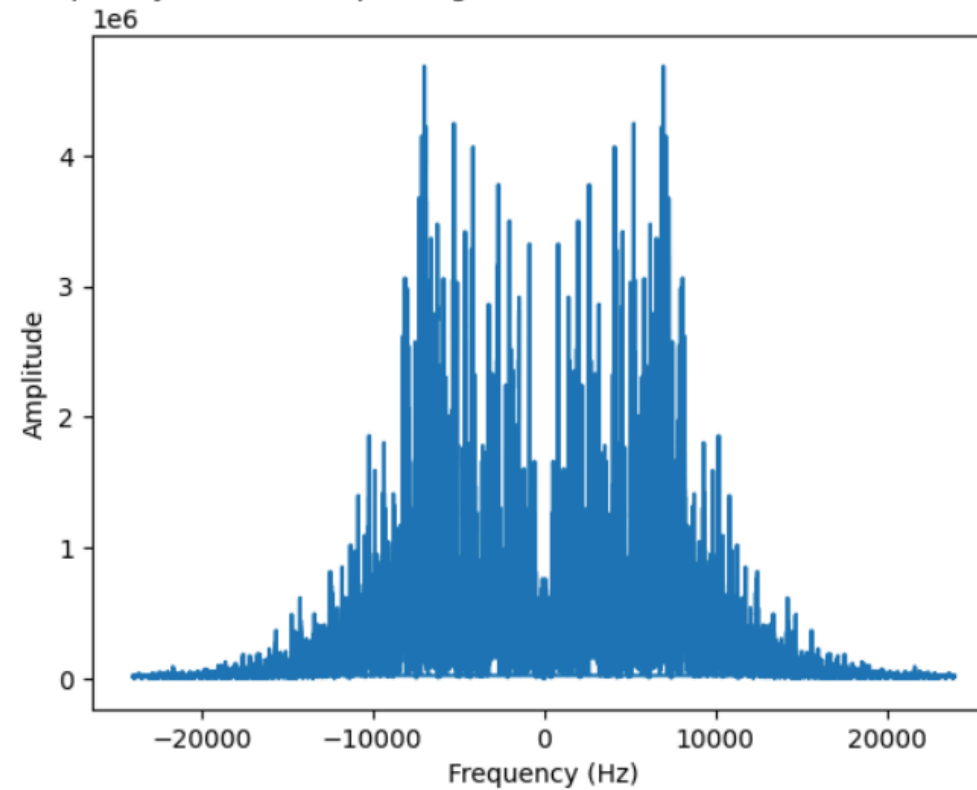
03 Nyquist frequency and aliasing

04 Exercises

| Negative frequencies

A two-sided spectrum

Frequency domain: <https://bigsoundbank.com/UPLOAD/wav/1234.wav>



Rotations in the complex plane

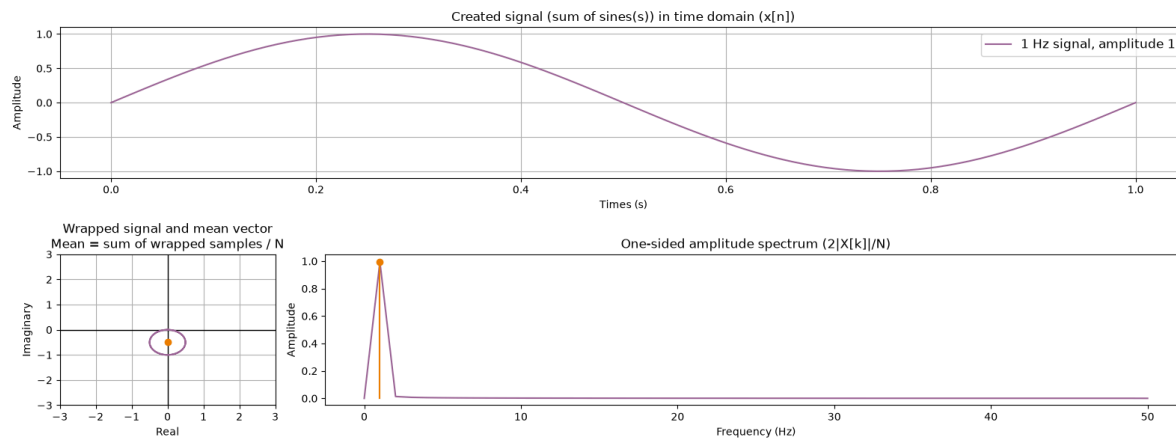


Source: <https://www.youtube.com/watch?v=SvFfVtRT1UI>

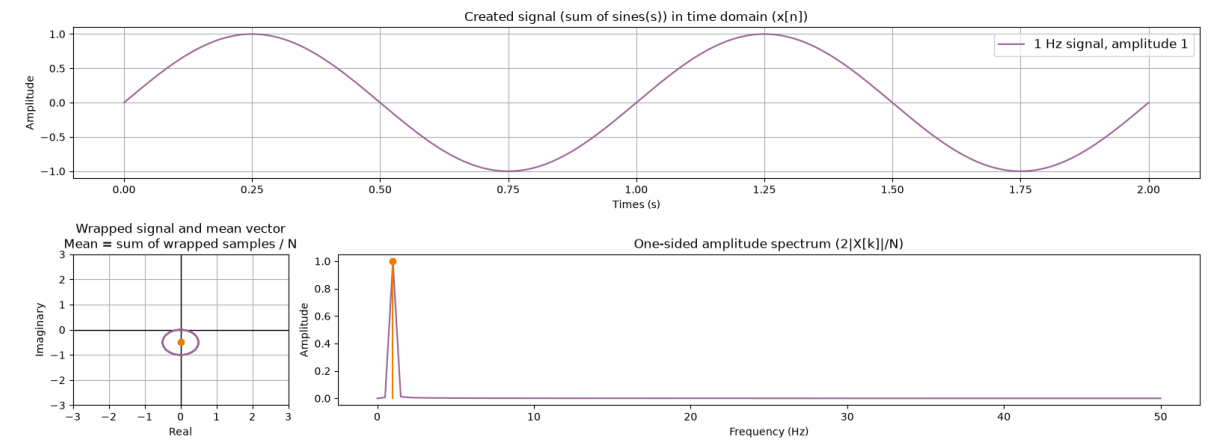
| Spectral leakage

One and two complete periods

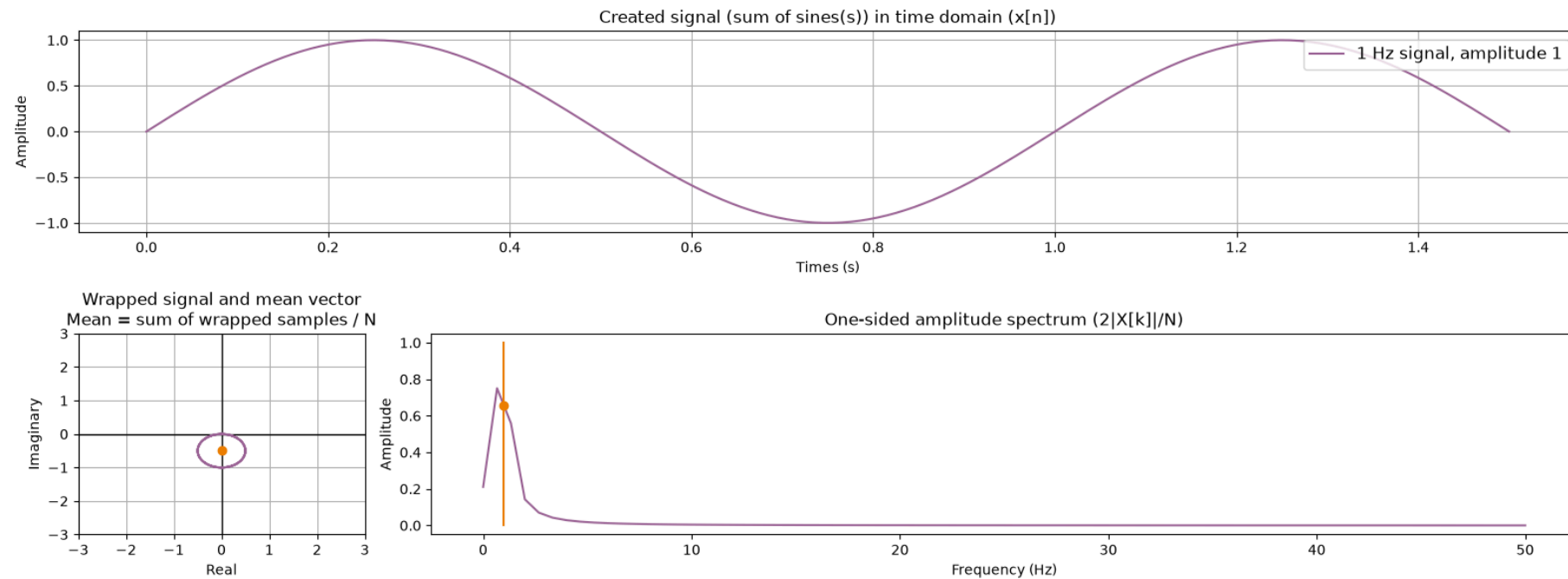
One complete period



Two complete periods

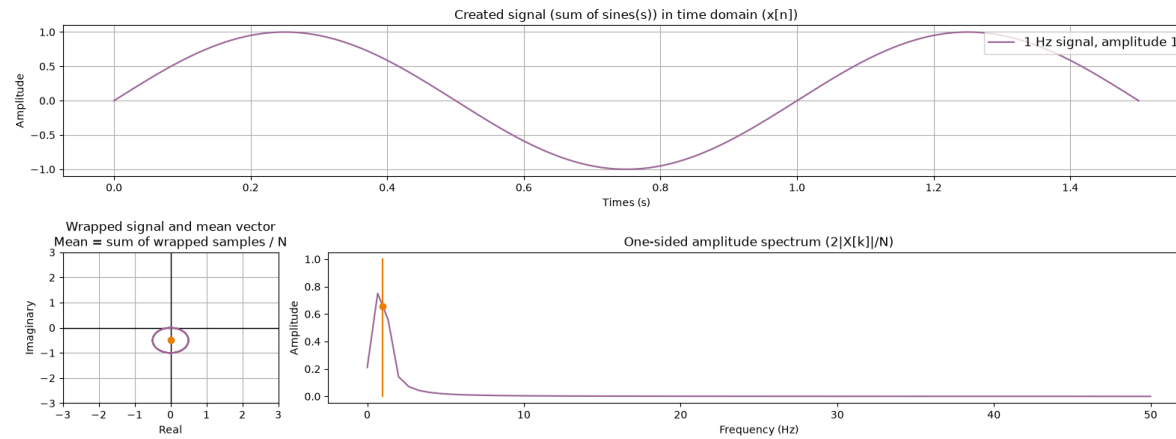


One and a half periods

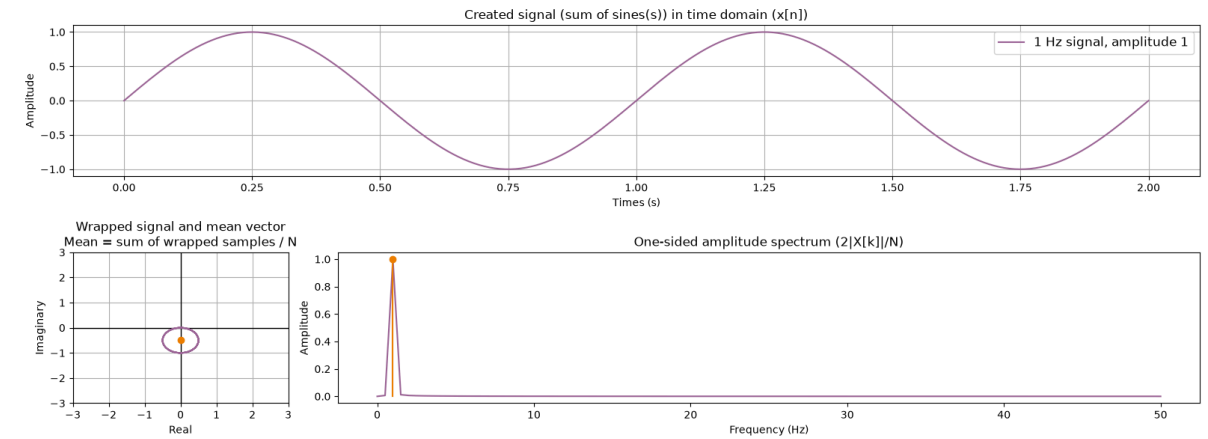


Comparing complete and partial periods

One and a half periods

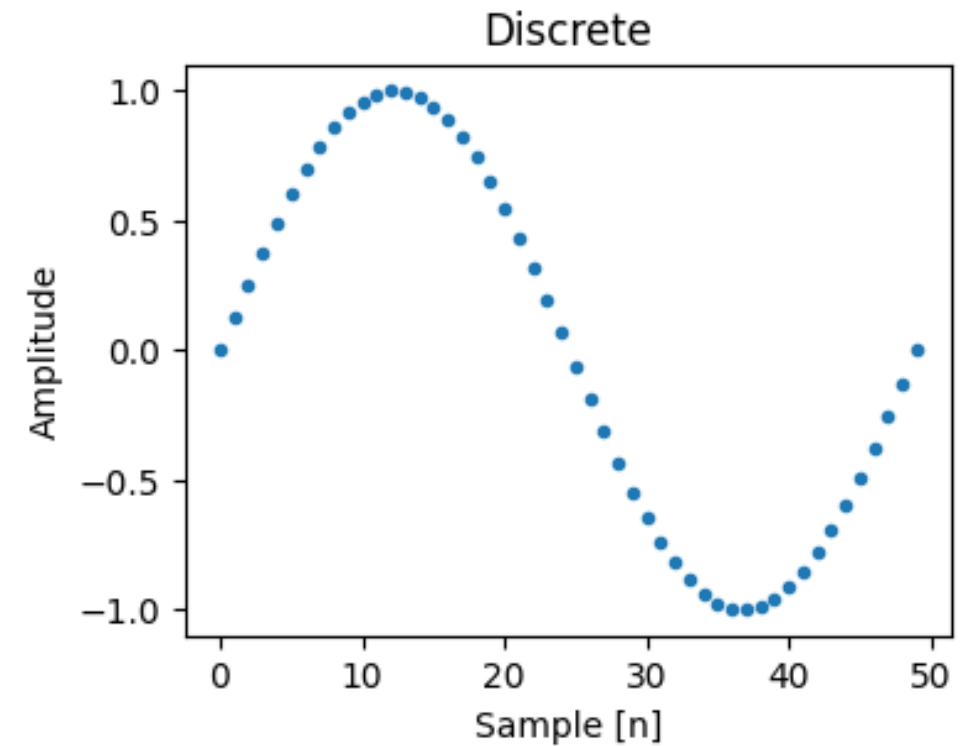
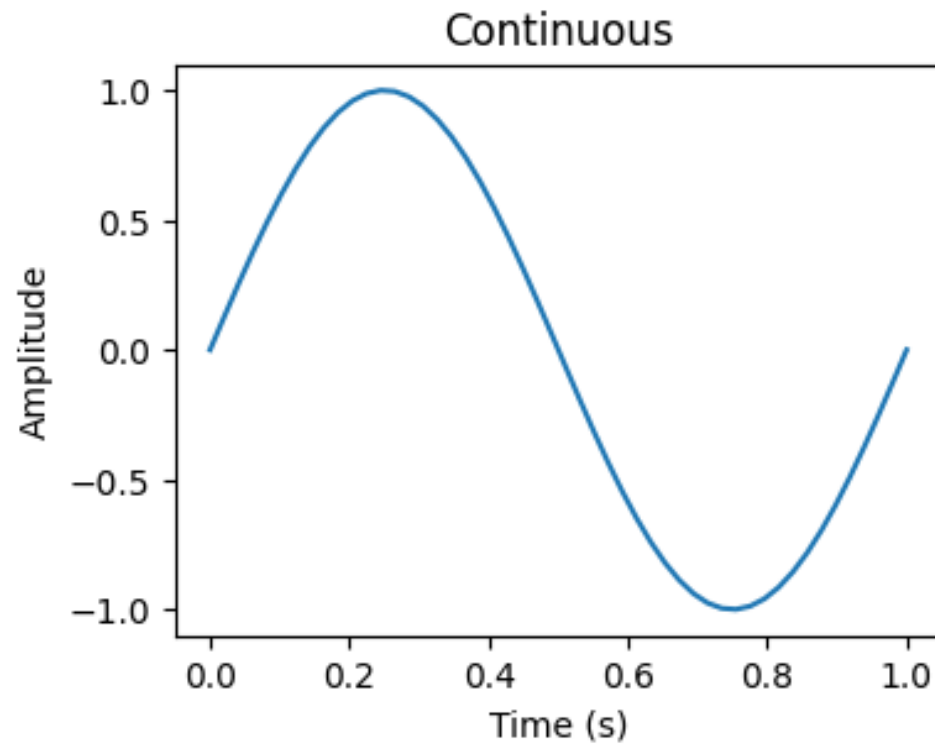


Two complete periods

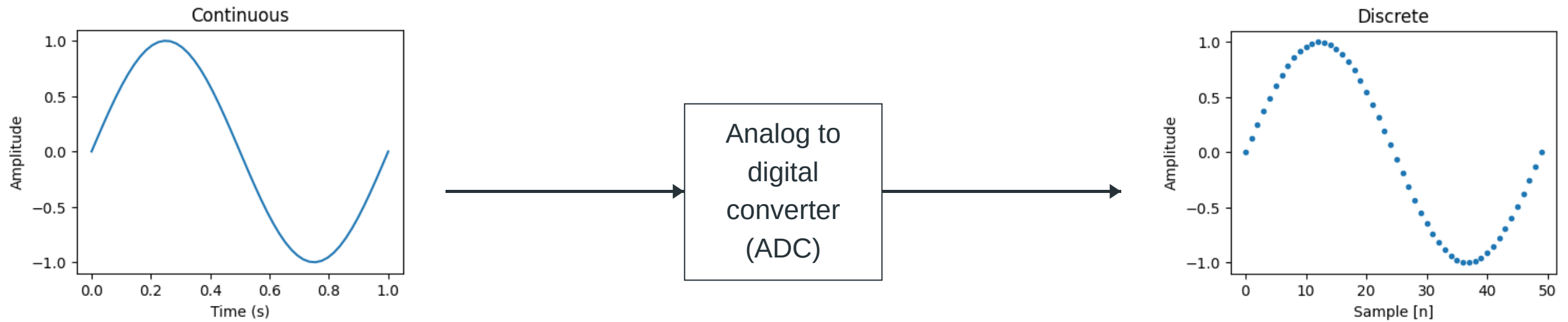


Nyquist frequency and aliasing

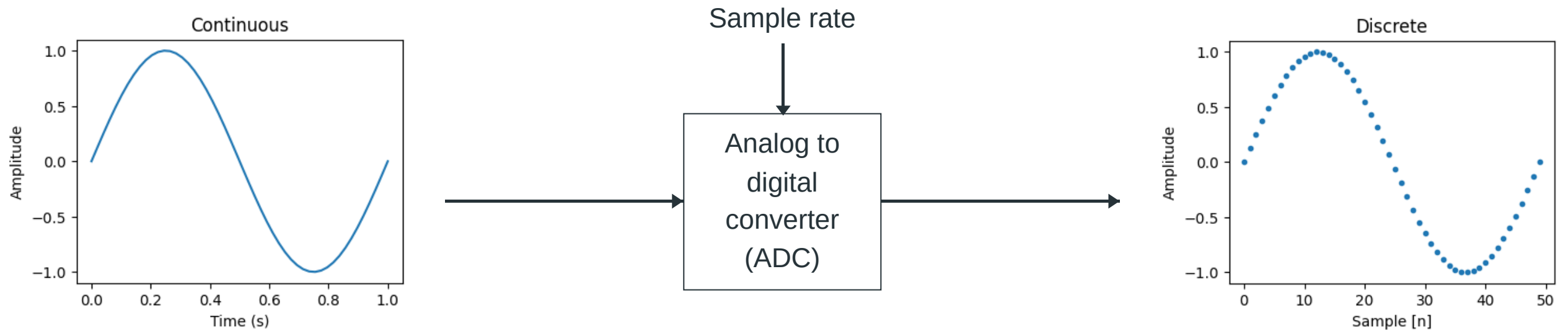
Continuous and discrete signals



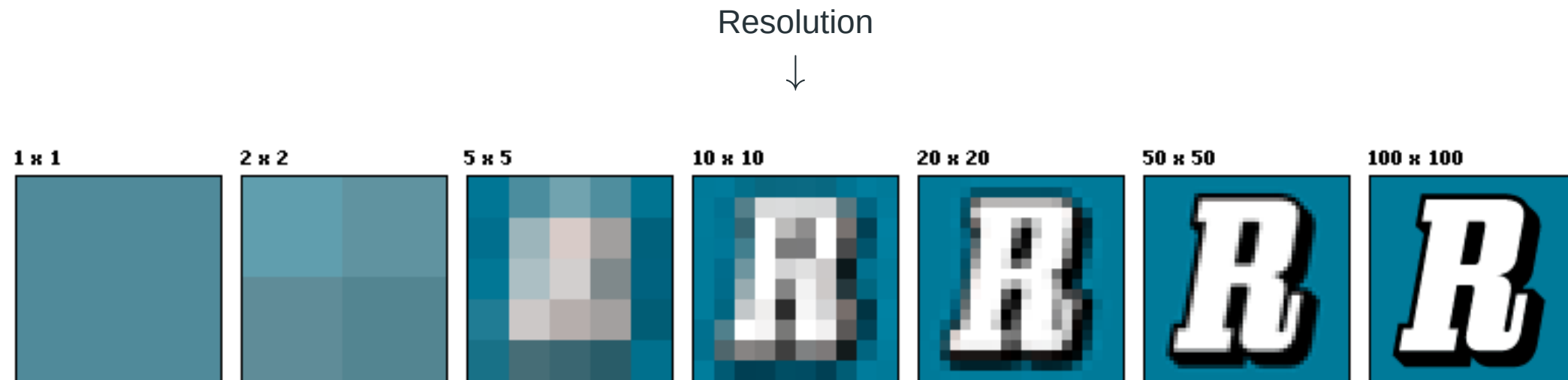
Analog to digital conversion



Sample rate

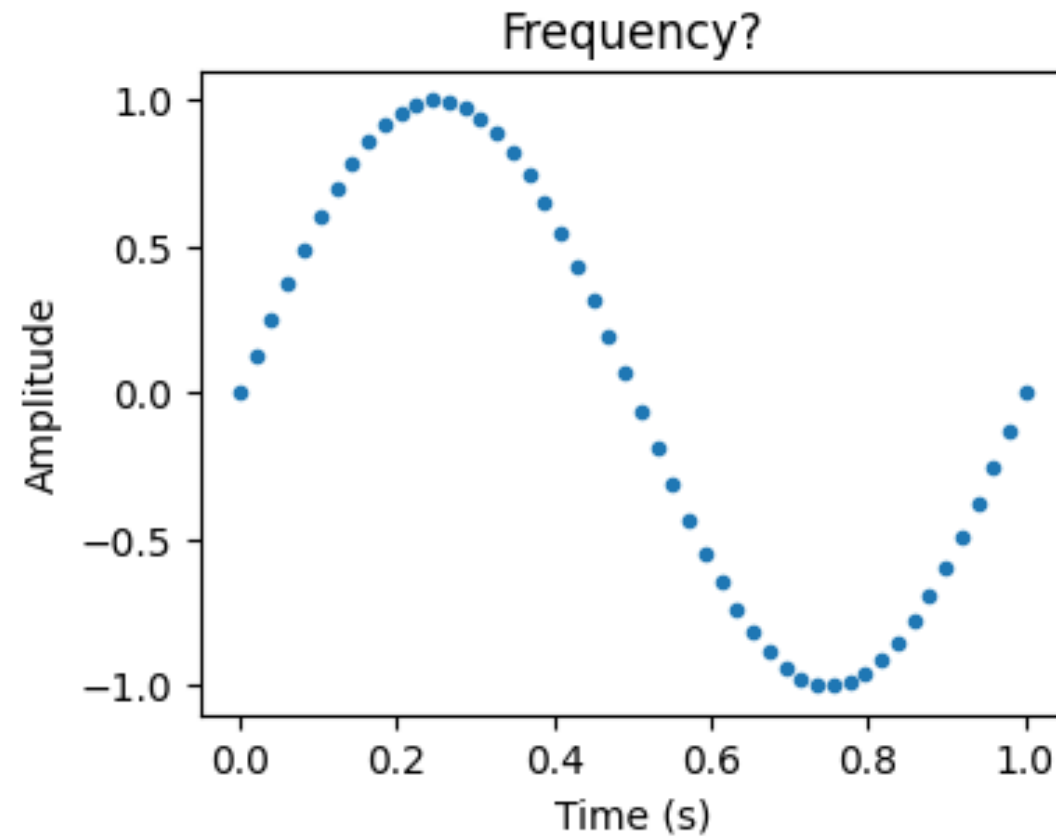


Resolution

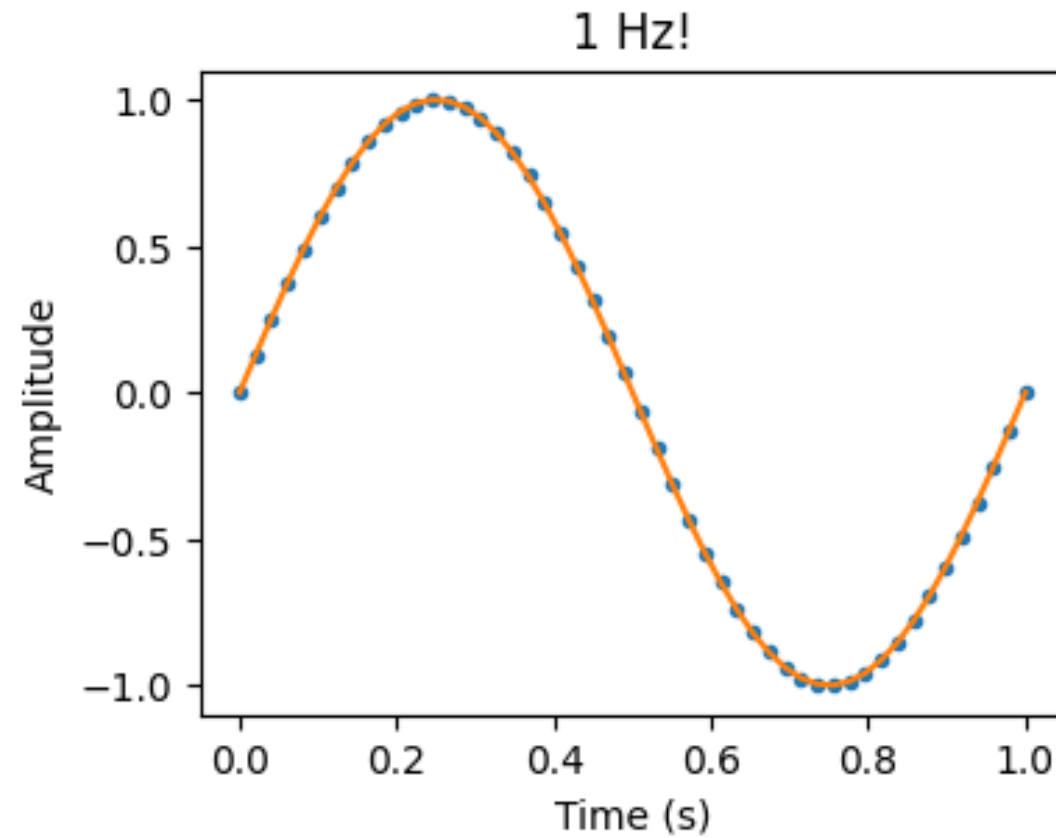


Source: https://commons.wikimedia.org/wiki/File:Resolution_illustration.png

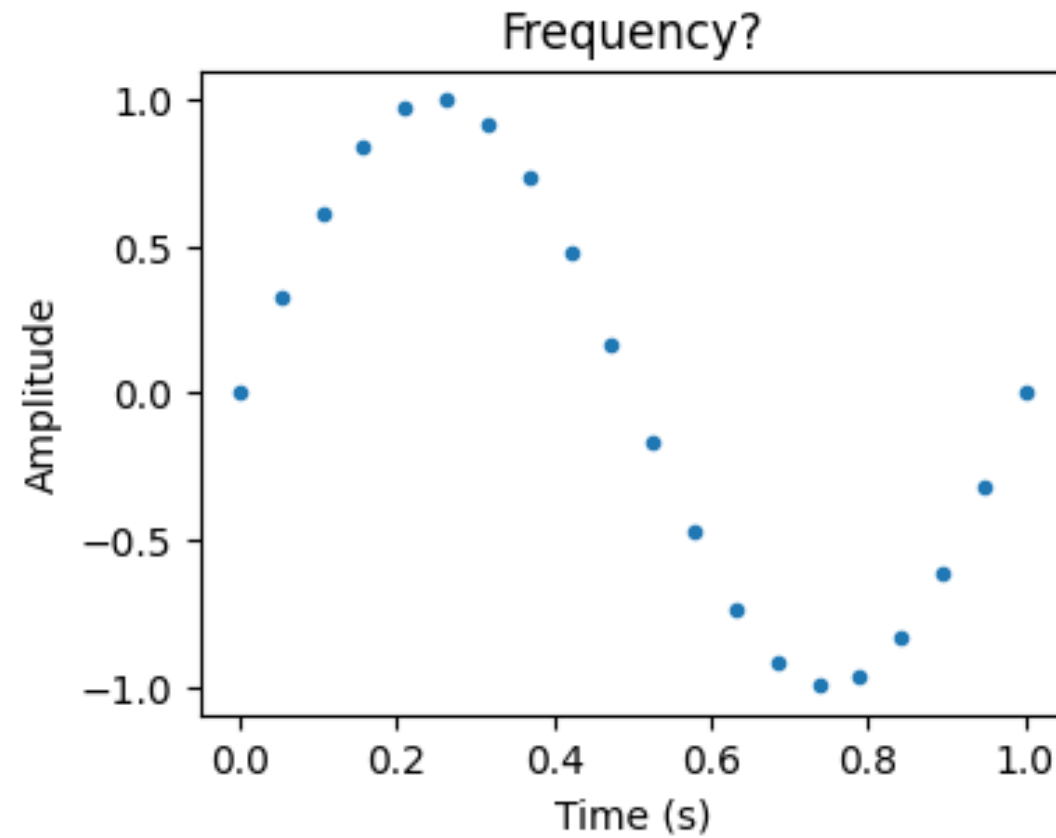
What is the frequency?



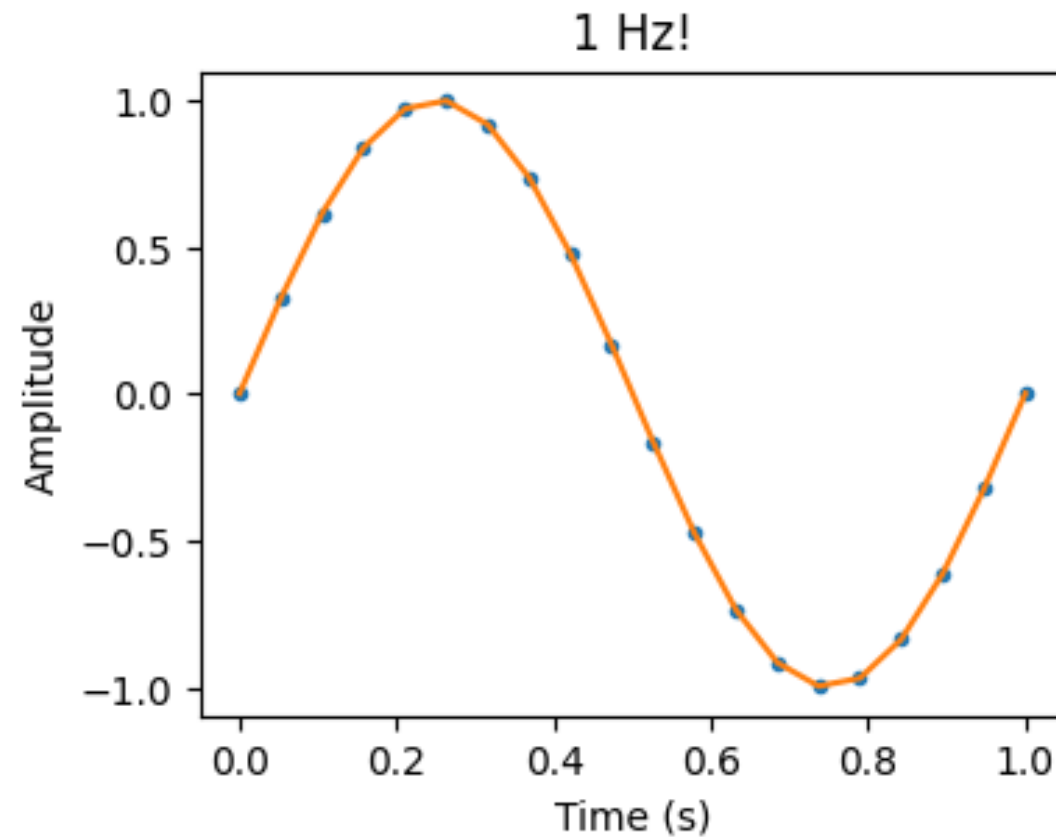
One cycle per second



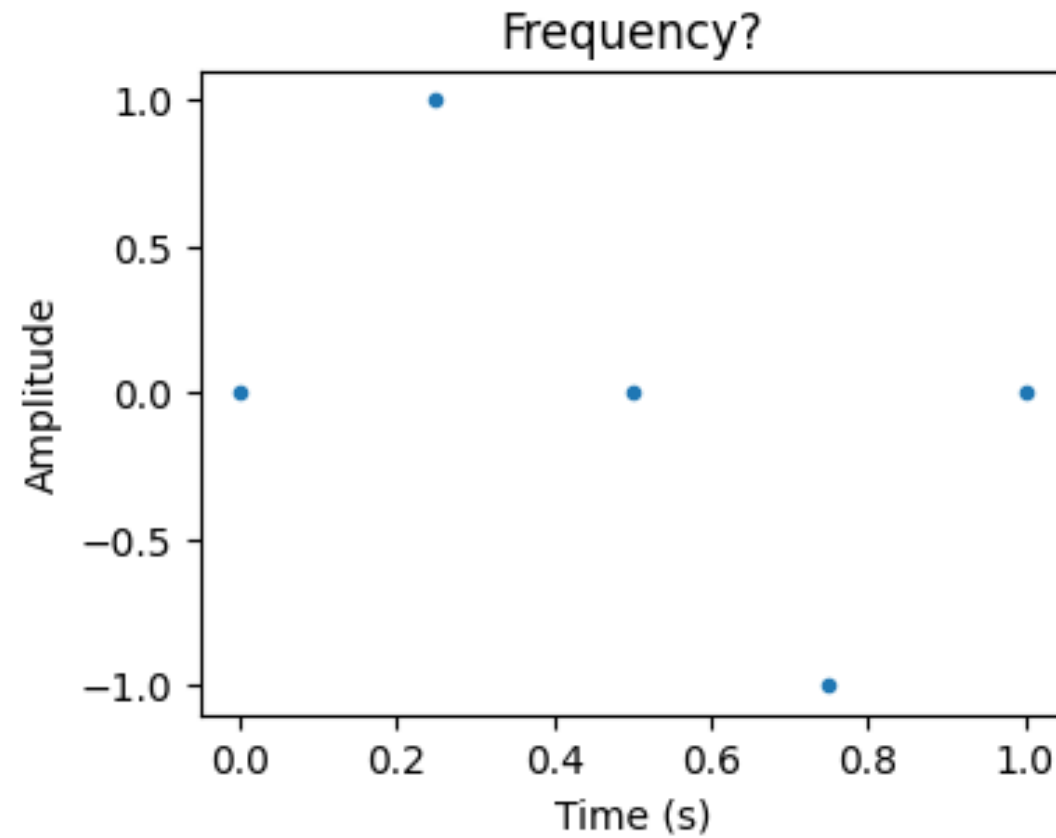
What is the frequency now?



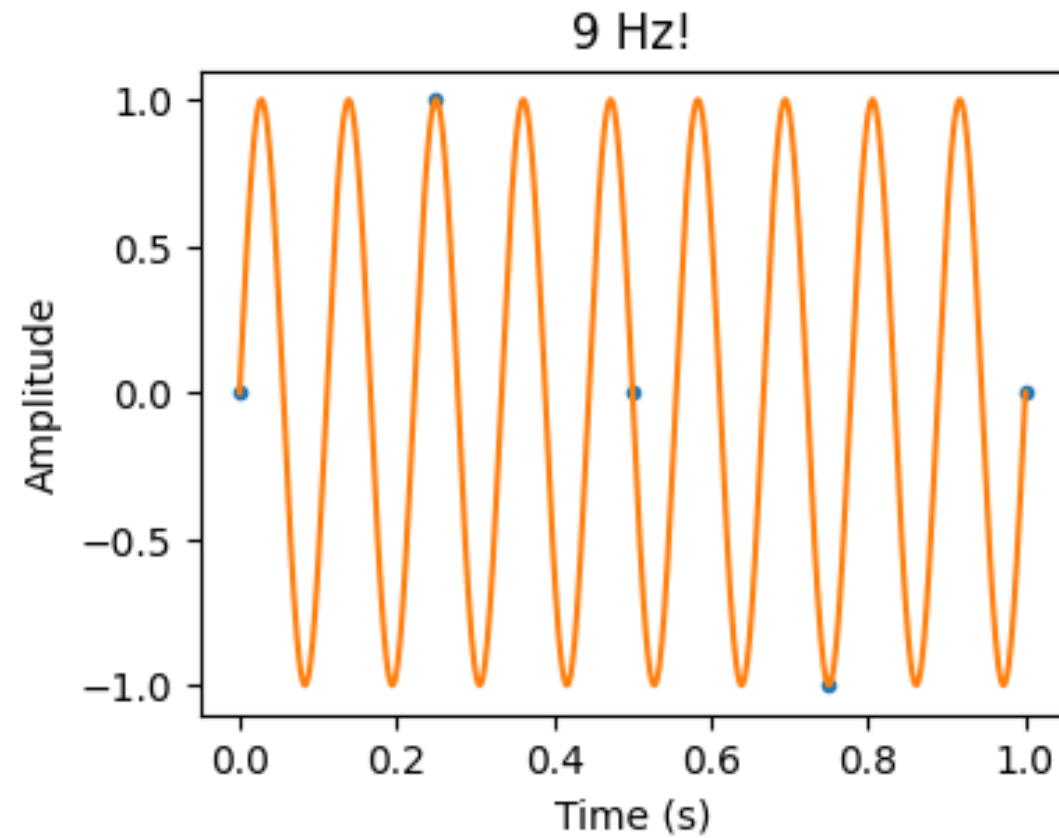
One cycle per second with fewer samples



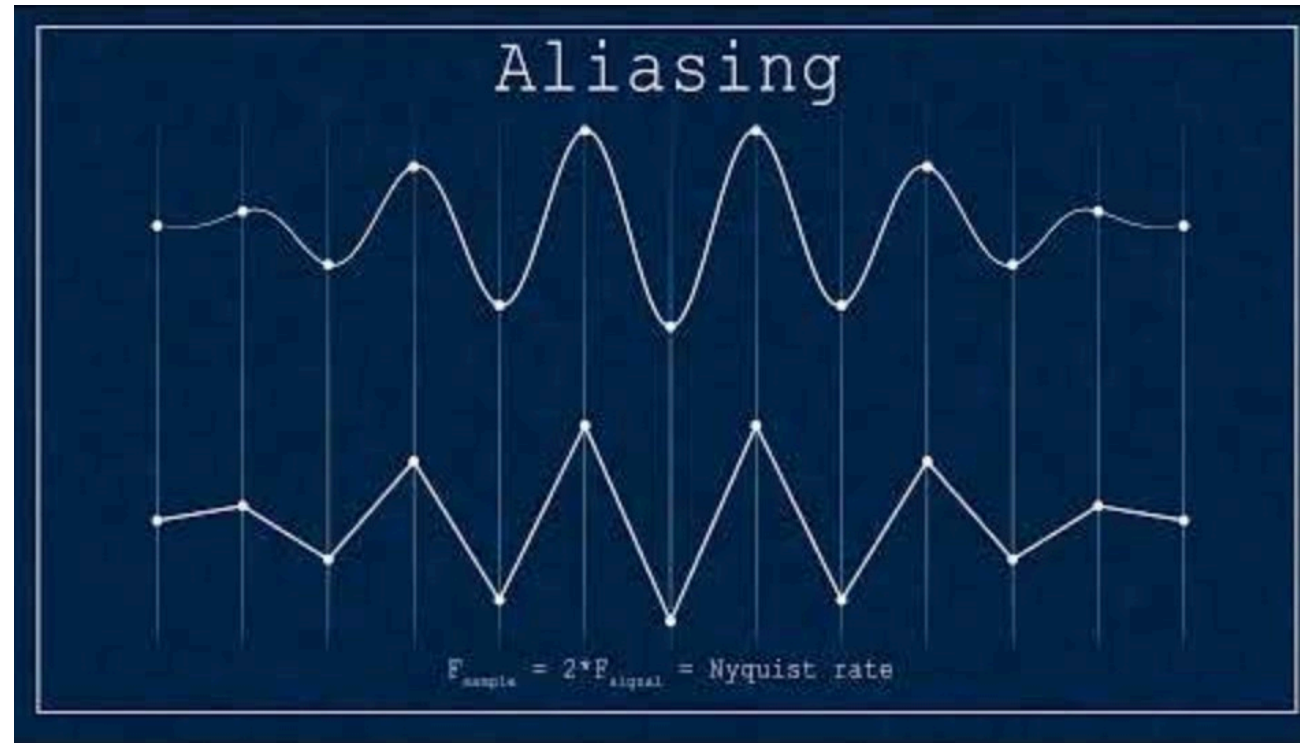
Can we distinguish these signals?



Aliasing



Aliasing in motion



Source: https://youtu.be/CTILCx5pHvM?si=TRD44vsEh_-Y7E3u

Nyquist frequency

$$f_{\text{Nyquist}} = \frac{1}{2} f_s$$

Half the sampling rate

The Nyquist frequency is half the sampling rate.

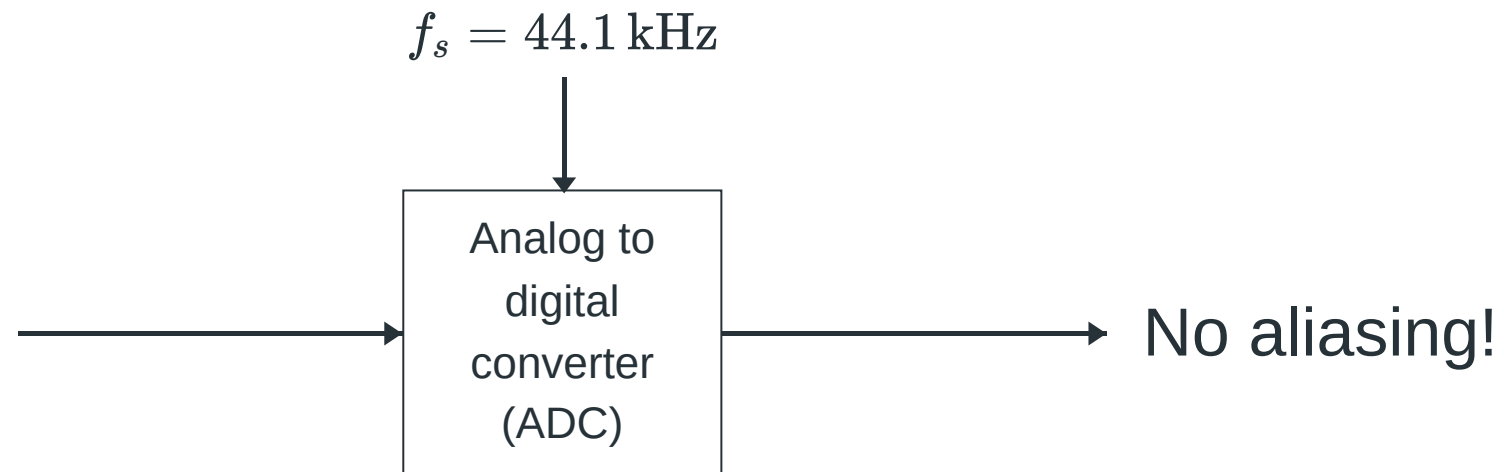
Why 44.1 kHz?

A common audio sample rate is 44.1 kHz, why?

We found 48 kHz previously, but that is beside the point.

What can we sample without aliasing?

?



Questions?

| Exercises

Exercises!

`../exercises/DSP-Lecture-04-Exercises.ipynb`